

DATA CLEANING, EDA, AND REGRESSION MODELING

2nd Homework Assignment

Due on: **May 13, 2026**

Introduction

This assignment focuses on data cleaning, exploratory data analysis, correlation analysis, multiple linear regression, model performance evaluation, regression visualization, residual diagnostics, and high leverage point detection using R.

You will work with a synthetic student-level dataset containing study hours, attendance rate, extracurricular activity participation, and final grades. The dataset intentionally includes missing values, inconsistent categorical labels, and logically impossible outliers so that you can practice realistic data preparation and modeling workflows.

Your goal is to produce a reproducible analysis that not only computes the requested statistics and models, but also interprets the results using sound statistical reasoning.

Learning Objectives

- Detect and handle missing values using an appropriate imputation strategy.
- Normalize inconsistent categorical values and convert them into a clean factor variable.
- Detect and correct outliers in numeric variables using defensible data-cleaning decisions.
- Perform exploratory data analysis using histograms and boxplots.
- Compute and interpret Pearson and Spearman correlation coefficients.
- Build and interpret a multiple linear regression model in R.
- Evaluate model performance using RMSE and residual diagnostics.
- Visualize regression relationships and identify influential observations using Cook's Distance.

Dataset

Use the following R code to create the assignment dataset:

```
set.seed(42)
df_assignment <- data.frame(
  Student_ID = 1:50,
  Study_Hours = c(10, 12, NA, 15, 8, 20, 22, 5, 30, NA, 14, 18, 45, 11, 9, 21, 25, 13, 16, 28,
    10, 12, 14, NA, 8, 19, 23, 6, 31, 15, 14, 18, 12, 11, 9, 22, 26, 13, 17, 29,
    10, 13, 15, 11, 7, 20, 24, 5, 32, 16),
  Attendance_Rate = c(0.85, 0.90, 0.70, NA, 0.60, 0.95, 0.88, 0.40, 1.0, 0.75, 0.82, NA, 1.20, 0.88, 0.65,
    0.92, 0.98, 0.76, 0.84, 0.99, 0.85, 0.91, 0.83, 0.68, 0.62, 0.96, 0.89, 0.42, 1.0, 0.77,
    0.81, 0.87, 0.84, 0.89, 0.66, 0.93, 0.97, 0.75, 0.85, 0.98, 0.86, 0.92, 0.82, 0.88, 0.61,
    0.94, 0.90, 0.41, 1.0, 0.78),
  Extracurricular = c("Yes", "no", "YES", "No", "no ", "Yes", "No", "no", "Yes", "YES", "No", "Yes", "Yes", "No", "no",
    "Yes", "No", "no", "Yes", "Yes", "No", "No", "Yes", "no", "No", "Yes", "No", "no", "Yes", "No",
    "No", "Yes", "No", "No", "no", "Yes", "No", "no", "Yes", "Yes", "No", "No", "No", "No",
    "Yes", "No", "no", "Yes", "No"),
  Final_Grade = c(75, 82, 60, 88, 55, 92, 85, 40, 98, 70, 78, 84, 250, 76, 62, 90, 95, 72, 80, 96,
    74, 83, 77, 65, 58, 91, 87, 42, 99, 71, 79, 85, 73, 75, 63, 92, 94, 70, 81, 97,
    76, 84, 78, 74, 56, 93, 88, 41, 100, 72)
)
```

Dataset Description

The dataset contains 50 student records and four analytical variables in addition to the student identifier. The table below summarizes the variables used in the assignment.

Column	Description	Expected Type	Data Quality Issues
Student_ID	Unique student identifier from 1 to 50.	integer	No intentional issue.
Study_Hours	Number of study hours.	numeric	Contains missing values and an extreme high value.
Attendance_Rate	Attendance rate represented as a proportion.	numeric	Contains missing values and a logically impossible value greater than 1.0.
Extracurricular	Participation in extracurricular activities.	categorical / factor	Contains inconsistent spellings and spacing.
Final_Grade	Final grade on a 0-100 scale.	numeric	Contains a logically impossible grade greater than 100.

Assignment Overview

Complete the following analytical tasks in R. Your response should include code, visualizations, model output, and short written interpretations.

Section	Topics Covered	Expected Output
Task 1	Data cleaning	Clean dataset and justify decisions
Task 2	Exploratory data analysis	Histogram and boxplot
Task 3	Correlation analysis	Pearson and Spearman coefficients
Task 4	Regression modeling	Multiple linear regression model
Task 5	Model performance	Manual RMSE calculation
Task 6	Interpretation	R-squared and coefficient interpretation
Task 7	Regression visualization	Scatter plot with regression line
Task 8	Residual analysis	Residuals vs fitted plot and assumptions
Bonus	High leverage points	Cook's Distance visualization and interpretation

Tasks

Task 1 — Data Cleaning

- **Missing Values:** Identify the NA values in the dataset. For continuous variables, apply median imputation and explain why the median is appropriate.
- **Inconsistency:** Normalize the inconsistent values in the Extracurricular column such as "Yes", "YES", "no ", and "no", and convert the column into a two-level factor.
- **Outliers:** Check whether the Final_Grade and Attendance_Rate columns contain logically impossible values, such as grades above 100 or attendance rates above 100%. Detect these outliers and correct them using an appropriate method such as deletion or winsorization.

Task 2 — Exploratory Data Analysis (EDA)

- Visualize the distribution of the Final_Grade variable using a histogram.
- Compare final grades by extracurricular activity participation using a boxplot.

Task 3 — Statistical Analysis: Correlation

- Calculate the Pearson and Spearman correlation coefficients between Study_Hours and Final_Grade.
- Interpret the difference between the two coefficients with respect to possible non-linear relationships in the data.

Task 4 — Regression Modeling

- Build a multiple linear regression model where Final_Grade is the dependent variable and all remaining variables are independent variables.
- Inspect the model summary and identify which variables are statistically significant at $p < 0.05$.

Task 5 — Model Performance and RMSE

- Find the fitted values predicted by the model using `fitted.values`.
- Manually calculate the Root Mean Squared Error (RMSE) between the observed values and the predicted values using the RMSE formula.

Task 6 — Interpretation

- Interpret the explanatory power of the model based on the R-squared value you obtained.
- Answer the following question using the regression coefficients: How does the final grade change when study hours increase by one unit?

Task 7 — Regression Visualization

- Create a scatter plot using `Study_Hours` on the X-axis and `Final_Grade` on the Y-axis.
- Add the regression line to the plot using `geom_smooth(method = "lm")` or `abline()`.
- Interpret the plot by discussing how the data points are distributed around the regression line, how strong the relationship appears to be, and how well the model represents the data.

Task 8 — Residual Analysis

- Calculate the residuals of your model.
- Create a Residuals vs Fitted plot with fitted values on the X-axis and residuals on the Y-axis.
- Interpret whether the residuals are randomly distributed around the zero line or whether they show a funnel shape or U-shaped pattern.
- Discuss what the plot indicates about the linearity assumption and constant variance assumption (homoscedasticity). State whether the model appears reliable.

Bonus Task — High Leverage Points

Investigate whether there are influential observations in the dataset that strongly affect the slope of the model. Visualize Cook's Distance values and interpret the result.

Submission Format

- Submit your work as a single R Markdown (.Rmd) file or a PDF output that includes all analysis steps.
- Include interpretations as `print()` outputs or code comments where appropriate.
- Submit a single R Markdown file named **student-id.Rmd**, replacing **student-id** with your **actual student ID**. Your file must include all R code, outputs, visualizations, and written interpretations.
- All visualizations must be complete and properly labeled with titles and X-Y axis labels.

Important Notes

- Your code must run from top to bottom without errors. Submit a single R Markdown file named **student-id.Rmd**, replacing **student-id** with your **actual student ID**. Your file must include all R code, outputs, visualizations, and written interpretations.
- Use clear and descriptive variable names.
- Every data-cleaning decision must be justified briefly.
- Graphs without titles or axis labels will be considered incomplete.
- Academic dishonesty includes but is not limited to copying code from classmates, sharing solutions, using AI-generated code, or any form of plagiarism — is strictly prohibited and subject to disciplinary action. Any student found guilty will receive an F grade for the entire course.
- Assignments submitted after the deadline will NOT be accepted.